higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

COMPUTER PROGRAMMING

(Second Paper)

NQF LEVEL 4

(10041024)

28 November 2019 (X-Paper)
09:00–13:00

This question paper consists of 13 pages.

Copyright reserved

TIME: 4 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.

2. Read ALL the questions carefully.

3. Number the answers according to the numbering system used in this question paper.

4. Students must be seated in the computer laboratory 30 minutes before the start of the examination to ensure that ALL computers have the appropriate programs.

5. Questions must be completed on a computer by using MS Office 2010 or a later version and VB.NET 2005 or VB.NET Express or VB.NET 2008 or a later version.

6. Print your work immediately after completing a question. Ensure that your EXAMINATION NUMBER appears on each PRINTOUT. NO questions without a printed EXAMINATION NUMBER will be marked.

7. In the event of problems such as a power failure or a computer or printer breakdown, the invigilator will make the necessary arrangements to continue with the examination process once the problem has been resolved. Students will NOT be penalised with time lost under these circumstances.

8. ALL work on the computer should be saved at regular intervals to prevent loss of work. NO additional time may be allowed for work lost due to lack of saving.

9. NO student may print his/her work for another student or make a memory stick available to another student or access another student's work on the network. Any attempt to access any information from or transfer information to another student in whatever manner is a contravention of the examination rules and regulations and will be viewed in an extremely serious light.

10. ALL PRINTOUTS or screen prints done during the session must be submitted.

Copyright reserved

11. Students may NOT copy the source code from VB.NET to MS Word and then print it from the word processor.

12. Steps to print a VB.NET form or any other screen:

    12.1 Run the solution.

    12.2 Hold the ALT key down and press the *Print Scrn* key which will copy the active screen.

    12.3 Open MS Word and paste the form/screen in a new document.

    12.4 Add your EXAMINATION NUMBER to the top right-hand side of the page.

    12.5 Print the page.

13. You are allowed to use the *Help* files in VB.NET.

14. The following files will appear on your hard drive in the following folders:

    14.1 QUESTION 1

    C:\CP L4 P2\FitAndFun\FitAndFun.sln
    (Microsoft Visual Studio Solution)

    14.2 QUESTION 2

    C:\CP L4 P2\Databases\Education.mdb
    C:\CP L4 P2\Learners_Extra_Classes\Learners_Extra_Classes.sln
    (Microsoft Visual Studio Solution)

    14.3 QUESTION 3

    C:\CP L4 P2\Images\know.jpg
    C:\CP L4 P2\Images\Style.css

15. Students may use the following programs or any other program of their choice to create webpages:

    - Notepad ++
    - Netbeans
    - Blue Voda
    - Dreamweaver

Copyright reserved

QUESTION 1

The following question refers to the VB.NET solution *FitAndFun* (CP L4 P2\FitAndFun) that has been saved on the hard drive.

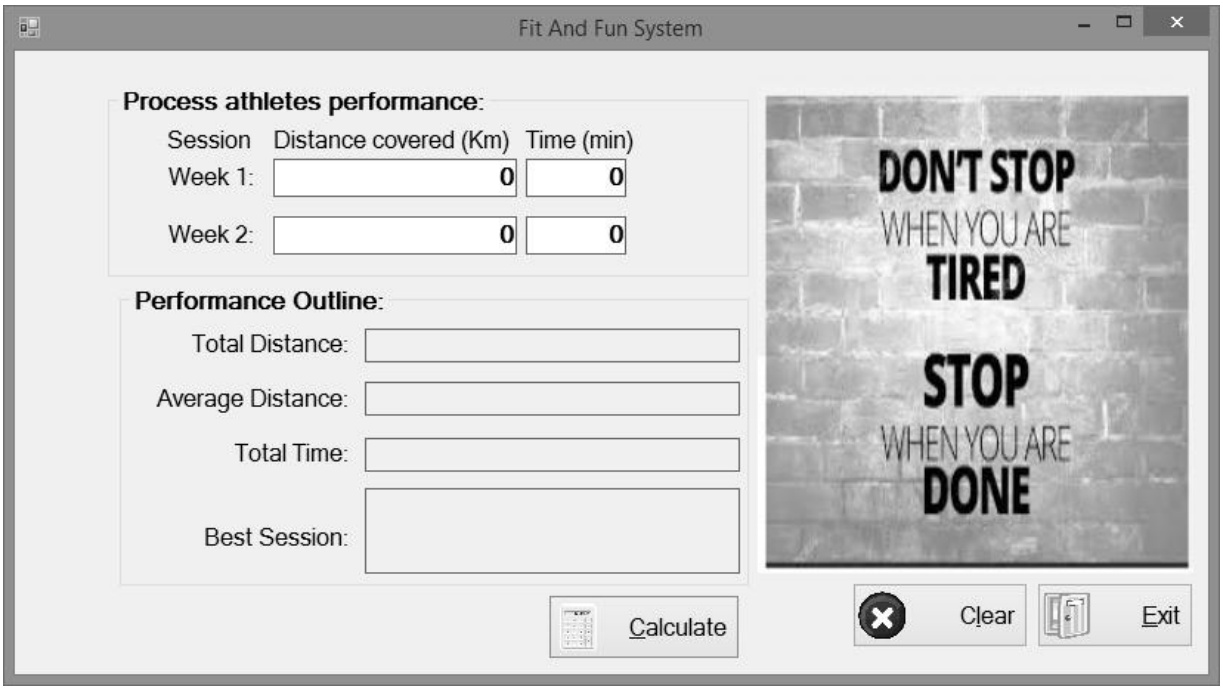
You have been tasked by members of the Fit And Fun organisation to create an application which will help them calculate the following items in a two week training session:

- Total distance
- Average distance
- Total time
- Best session

The program must capture the data of week 1 and week 2 as distance in kilometre and time in minutes respectively and complete the calculations as per the IPO table below.

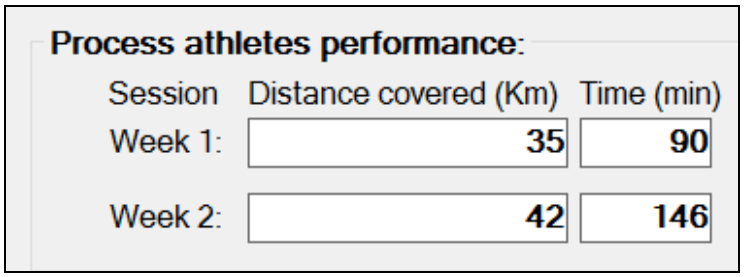
INPUT	PROCESSING	OUTPUT
distance1 time1 distance2 time2	totalDistance = distance1 + distance2 aveDistance = totalDistance/2 totalTime = time1 + time2 kilometerPerMinute1 = distance1/time1 kilometerPerMinute2 = distance2/time2	totalDistance aveDistance totalTime BestSession

Below is a print screen of what the user interface looks like.

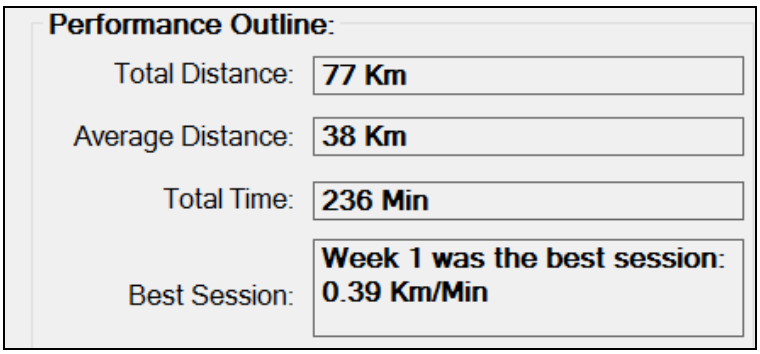


The following steps explain how to use the program to calculate and display the athlete's performance for TWO sessions.

STEP 1: Run the program and enter the distance1, time1 and distance2, time2.



STEP 2: Click on the *Calculate* button to calculate and display the athlete's run rate and best session.



Study the class diagram for the *Athlete* class and answer the questions.

Copyright reserved

The completed interface program has been saved on the hard drive. Open the solution *FitAndFun* and follow the instructions to complete the program.

1.1 Add the following program code to the *clsCalculateSession*:

1.1.1 Create the private fields. (1)

1.1.2 Create a parametrised constructor named *New* that will receive the values to be assigned to ALL fields as value parameters. The constructor should assign the passed values to the applicable fields. (2)

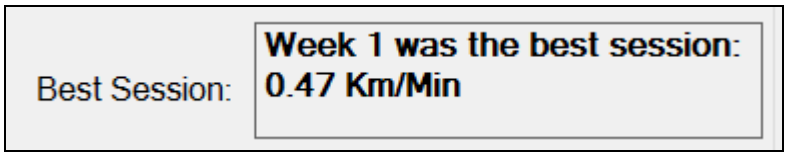
1.1.3 Create a public property for ALL the fields. (2)

1.1.4 Create the public function *CalDistance()* to calculate the distance covered by the athlete. (2)

1.1.5 Create the public function *calAverageDistance()* to calculate the average distance covered by the athlete. (2)

1.1.6 Create the public function *calTotalTime()* to calculate the total time covered by the athlete. (2)

1.1.7 Create the public function *BestSession()* to determine which week was the best and return a concatenated string as shown in the figure below. The string should be assigned to the label *lblBestSession*.



(3)

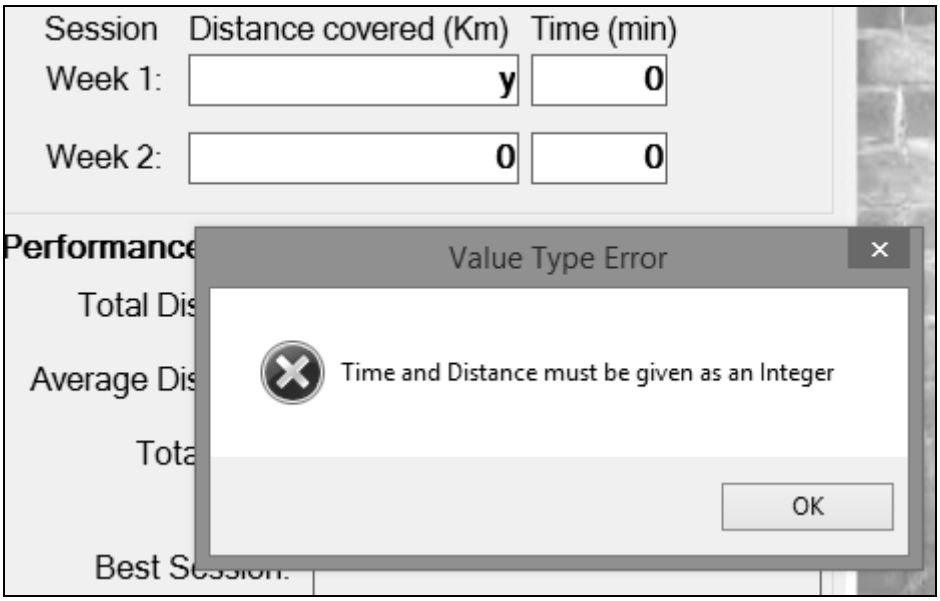
Add your EXAMINATION NUMBER to the program code of the *Athlete* class. Make a PRINTOUT of the program code of the class.

1.2 Add the following program code to the click event of the *Calculate* button on the athlete's form.

1.2.1 Write the program code to instantiate an object from the *Athlete* class. (1)

Copyright reserved

1.2.2 Write the required code to check if the textboxes contain numeric values **greater than zero** and display the error-message box shown below. On error the program should reset the textboxes and move the cursor focus to Week 1 distance (*txtDistanceWk1*).



(2)

1.2.3 Add code to the *Calculate* button *(btnCalculate)* that will send parameters to the *New* constructor to instantiate the *FitnessCalc* object. (5)

1.2.4 Display the results in the following labels:

- lblBestSession
- lblTotDistance
- lblAvgDist
- lblTotalTime

(2)

1.3 Change the text in the title bar of the form to your EXAMINATION NUMBER.

Run the program and the test data below. Calculate the athlete's performance and make a PRINTOUT of the form.

Week 1:	Distance	42 km
	Time	160 min
Week 2:	Distance	56 km
	Time	180 min

(1)
[25]

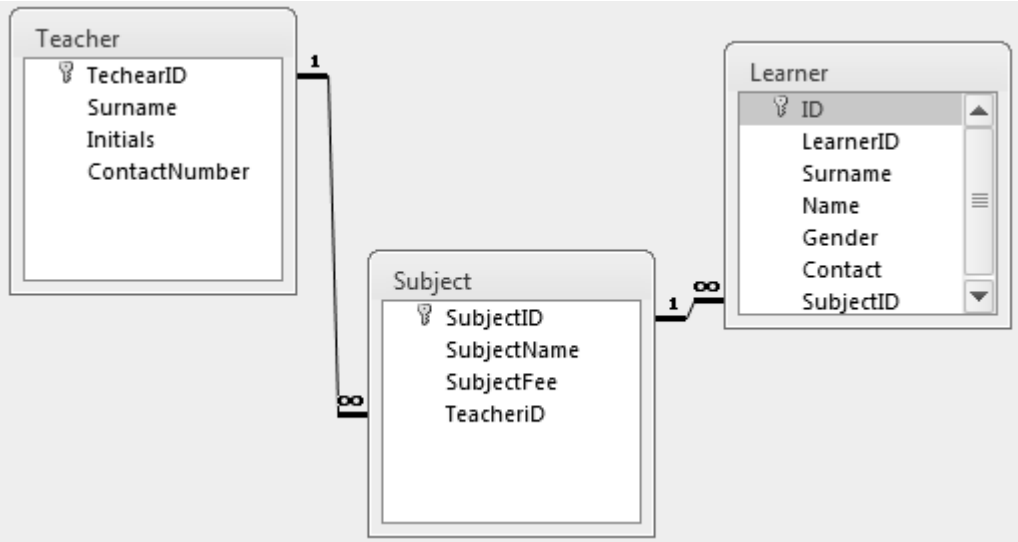
QUESTION 2

The following question refers to the VB.NET solution *Learners_Extra_Clasess* (CP L4 P2/Learners_Extra_Classes) that has been saved on the hard drive.

You must create an application for Better High School which will enable the school administrator to register learners who need extra classes and view the learner register list.

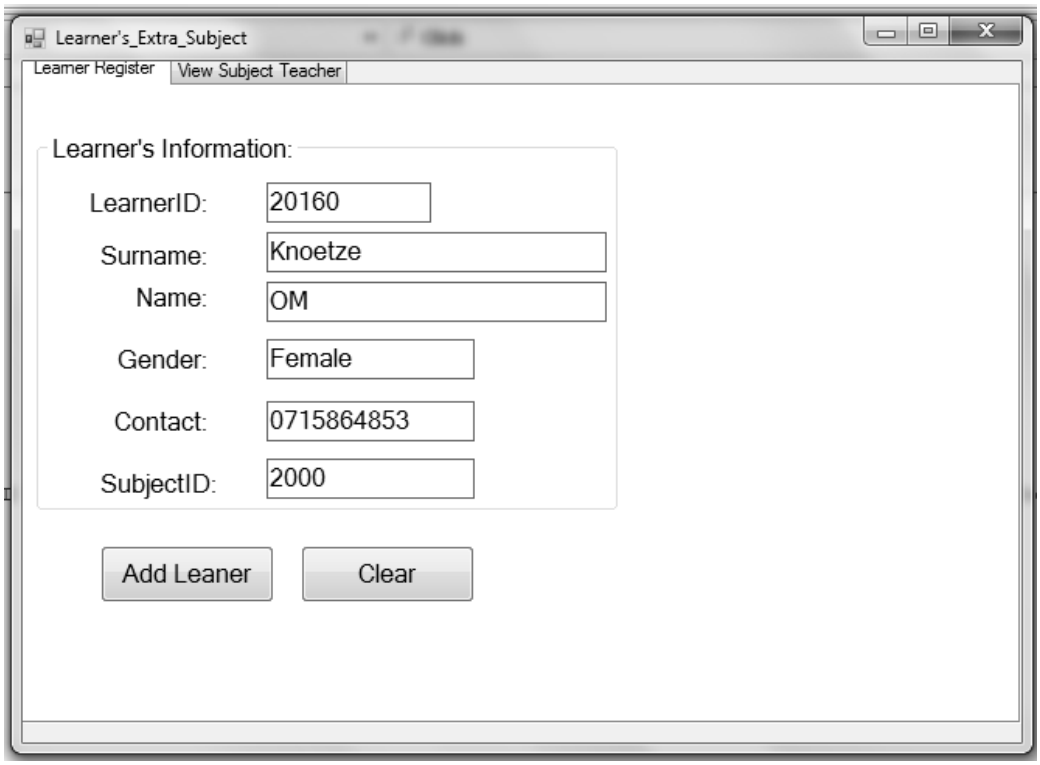
A database called *Education.mdb* with related data-tables *Teacher*, *Subject* and *Learner* has already been created and populated with data. It is located in the folder called C:\CP L4 P2\Learners_Extra_Classes.

Below is the graphical representation of the three tables (*Teacher*, *Subject*, *Learner*) and the existing relationships between them.

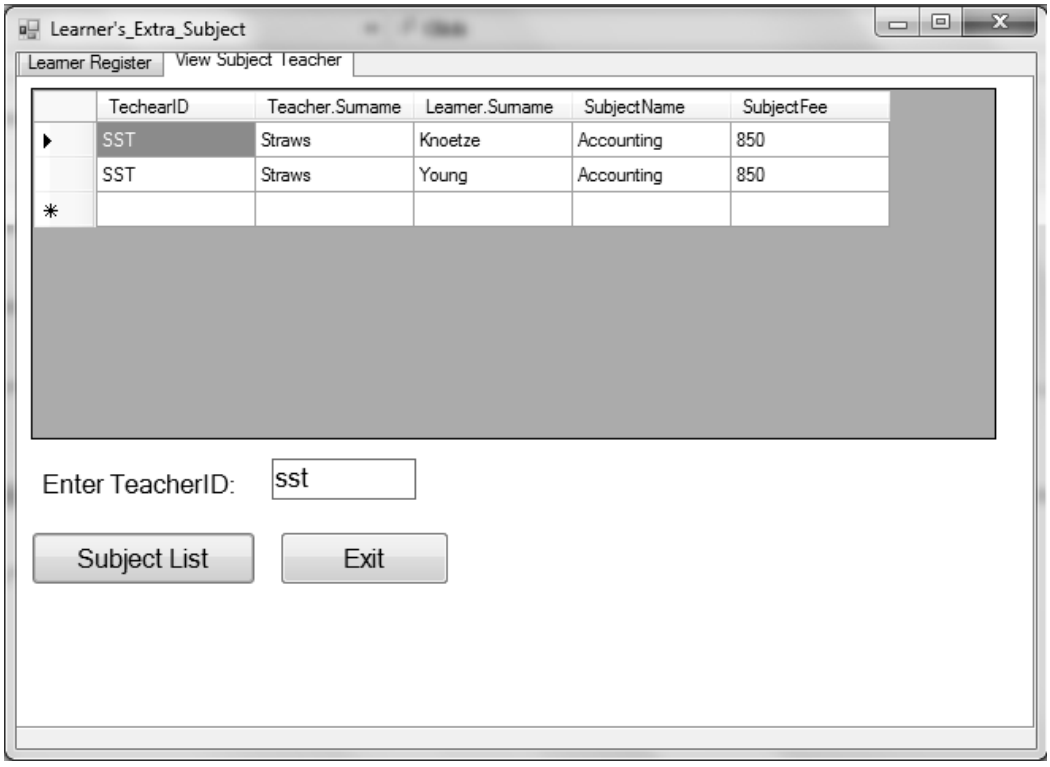


The steps below explain how to use the program to register a learner's extra subject as well as how to view the teacher-subject list from the database.

STEP 1: Run the program. Enter the learner's information and click the *Add Learner* button on the *Learner Register* tab page.



STEP 2: Once the record has been added into the table *Learner*, click on the *View Subject Teacher* tab page to retrieve records with *TeacherID*.



Copyright reserved

2.1 Add the required Visual Basic code to the click event of the *Add Learner* button on the *Learner Register* tab page that will:

2.1.1 Establish a connection to the database. (4)

2.1.2 Insert the record into the *Learner* database table. (8)

2.2 On the *View Subject Teacher* tab page, add the required Visual Basic code to the click event of the *Subject List* button to:

2.2.1 Establish a connection to the database. (4)

2.2.2 Retrieve the *TeacherID*, *teacherSurname*, *learnerSurname*, *SubjectName* and *SubjectFee* from the three related tables for all the records that match the value of the *TeacherID* that was entered. (6)

2.2.3 Display the retrieved records in the data grid viewer. (1)

2.3 Change the text in the title bar of the form to your EXAMINATION NUMBER.

Run the program with the following test data and take screen shots of the results.

On the *View Register* tab page enter the following data in the designated field elements.

LearnerID	20176
Surname	Sunrise
Name	Lucia
Gender	Female
SubjectID	1000

(1)

On the *View Subject Teacher* tab page retrieve the record values for the following *TeacherID*:

TeacherID PJS (1)

Make a PRINTOUT of the source code and include the screenshots of the results. **[25]**

QUESTION 3

The image in FIGURE 1 is a screenshot of a webpage belonging to 'SK College'. It allows students to calculate the cost of registration based on their selected *Major Subject* and whether or not they *Select Residence* on or off campus. You are required to reproduce the page as shown by answering the questions below:

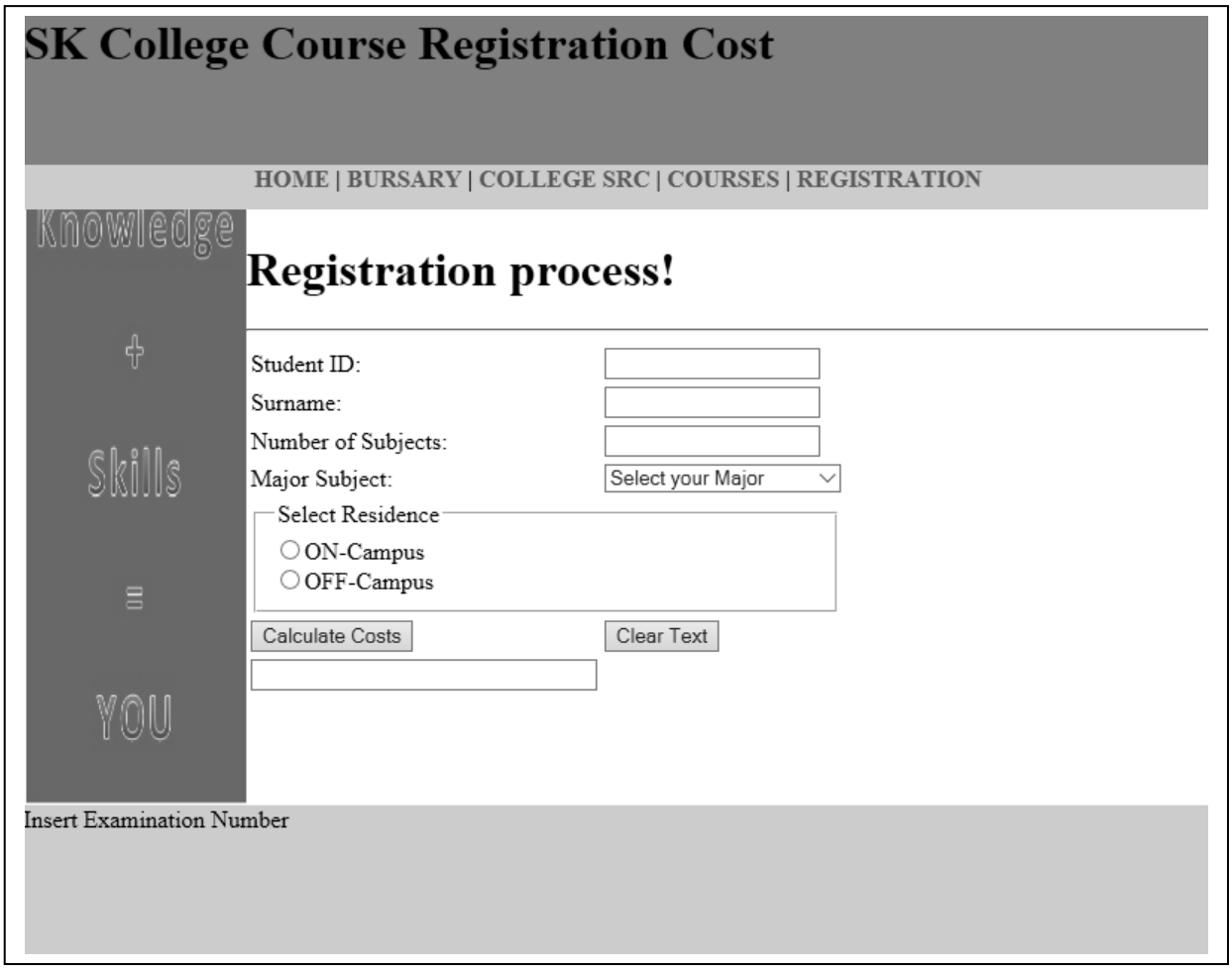


FIGURE 1 (*SK College Course Registration Cost)*

3.1 Use the editor of your choice and open the external style sheet *Style.css* located in C:\CP L4 P2\Images\Style.css. Insert your EXAMINATION NUMBER as a comment in the style sheet.

3.1.1 Create a rule set for a normal, unvisited link and set the text decoration attribute to none. (2)

3.1.2 Create a rule set for the navigation division. Set the height to 30 px, text alignment to center, font weight to bold and background colour to #ccc. (3)

3.2 Create a new HTML file with ALL the required divisions and save it as *index.html.* Answer the following questions to complete the necessary HTML code that will the replicate the webpage.

3.2.1 Add a link to the external cascading style sheet called *Style.css*. (1)

3.2.2 Create the header division. Insert the heading 'SK College Course Registration Cost' as a h1 heading element. (1)

3.2.3 Create the navigation division. Insert the menu hyperlinks HOME, BURSARY, COLLEGE SRC, COURSES and REGISTRATION. Set the hyperlink reference property of each hyperlink to *Home.html*, *Bursary.html*, *CollegeSRC.html, Courses.html* and *Registration.html,* respectively. (5)

3.2.4 Create the sidebar division. Insert the image *know.jpg*, located in the *Images* folder and set the width property to 150 and the height property to 400. (2)

3.2.5 Create a borderless table element with FIVE rows and TWO columns inside the main division of the webpage. (1)

3.2.6 Create a form element called *register*. (1)

3.2.7 Insert the text, 'Registration Process' as a h2 heading element followed by a horizontal rule. (2)

3.2.8 Add the form input elements for *Student ID*, *Surname* and *Number of Subjects*. (3)

3.2.9 Create the form dropdown list for *Major Subject* and populate it with the options *Select your Major, Software Development, Networking, Analyst* and *Finance*. (3)

3.2.10 Create a field set with the title 'Select Residence' to group the radio button options for *ON-Campus* and *OFF-Campus* as shown in FIGURE 1. (3)

3.2.11 Add the button *Calculate Costs* with an 'on click' event handler called *compute* to calculate the total course costs. (2)

3.2.12 Add the button *Clear Text* to clear fields for data entry. (1)

3.2.13 Add a form element *regCost* that will display the registration cost below the *Calculate Cost* and *Clear Text* buttons (1)

3.2.14 Make a PRINTOUT of the source code with a SCREENSHOT of the page as displayed in your browser. (1)

[32]

QUESTION 4

This question refers to the webpage (FIGURE 1) in QUESTION 3 above. Use your knowledge of JavaScript to answer the questions below.

You are required to add the functionality to the webpage that will calculate and display the *Registration Cost* when the user clicks the *Calculate Cost* button.

4.1 Add a script element to the webpage and indicate that JavaScript will be used.

4.1.1 Create a JavaScript function called *compute()* which will be called from the 'onclick' event of the *Calculate Cost* button. (1)

4.1.2 Declare the variables *nrSubject, campus* and *totalCost* (3)

4.1.3 Assign the form values of *Number of Subjects* and *Select Campus* to the variables *nrSubject* and *campus*, respectively. (2)

4.1.4 Validate the *Number of Subjects* field to ensure it is numeric and NOT empty. (3)

4.2 Use an *if* statement to determine which checkbox has been checked and then execute the calculation as indicated.

4.2.1 If the *ON-Campus* radio button is checked, calculate the total course cost as:
Total cost = number of subjects multiplied by the unit price of R750,00 plus the residence fee of R8 500,00. (4)

4.2.2 If the *OFF-Campus* radio button is checked, calculate the total course cost as:
Total cost = the number of subjects multiplied by the unit price of R750,00. (2)

4.2.3 Display the value of *Total Cost* as a currency value in the *regCost* form element. (2)

4.2.4 Make a PRINTOUT or take a SCREENSHOT of the browsed webpage showing the result of the calculation. (1)

[18]

TOTAL: 100

Copyright reserved